

$Re_Q = d^2 \omega / \nu$ at $Re_\theta = Re_{\theta_c}(H)$

- $\text{softmin}_2(2600, 175 + 2(\hat{S}g)^{-2})$
- favorable $\beta = +0.15$ ($H=2.44, Re_{\theta_c}=1235$)
- Blasius $\beta = 0$ ($H=2.59, Re_{\theta_c}=242$)
- adverse $\beta = -0.10$ ($H=2.80, Re_{\theta_c}=97$)
- incipient separation ($H=3.98, Re_{\theta_c}=36$)
- separated (reversed flow) ($H=10.63, Re_{\theta_c}=13$)

$\hat{S}g$

10^{-2}

10^{-1}

10^0

10^3

10^2